

DISTRIBUTION OF RUNS TESTS UNDER VARIOUS MODELS

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Key words: Bernoulli trials; learning experiments; null distribution; power of test; randomness.

Summary: Using the results of Fu and Koutras (1994), an investigation is undertaken into the distribution of a class of runs tests under the hypothesis of randomness as well as a number of nonrandomness hypotheses. Power comparisons of members of the class show some unexpected characteristics such as test bias and lack of monotonicity in some cases. As an application it is then shown how one can go about deciding on the best member of the class in psychological learning experiments.

1. INTRODUCTION AND BACKGROUND

Runs tests are used in various fields: Reliability Engineering, Quality Control, Psychology and many others. In Psychology, for example, the theory that success breeds success and/or failure breeds failure, is often considered in achievement testing. In animal learning experiments, a test animal performs a sequence of trials and in each trial a box containing food is either selected or not, producing a series of trials (see Schwager, 1983).

In this paper we study the application of runs tests to detect various forms of nonrandomness. As is commonly known, test statistics used in runs tests exist in various forms, such as

the number of success runs of size exactly k , $k = 1, 2, \dots$;

the number of success runs of size greater than or equal to k , $k = 1, 2, \dots$; and

the number of nonoverlapping consecutive k successes.

The major difficulty with these tests has always been that the null distributions of the test statistics were conditional on the number of successes and the number of failures. Furthermore, finding distributions under alternative hypotheses was difficult if not impossible.

In a very elegant way Fu and Koutras (1994) produced a broad framework within which the distribution of many of these test statistics can be studied. This approach enables one to study the unconditional distribution of runs statistics, both under the null and alternative hypotheses. Critical values of tests, power comparisons and sample size determination have now become relatively easy to study.

2. BRIEF OVERVIEW OF PRESENT STUDY

In this study we first investigate the null distribution of a class of runs statistics, namely $G_{n,k}$, the number of success runs of length at least k , $k = 1, 2$, and 3 . General properties of the null distribution of the statistics are discussed.

Secondly we calculate the power of the tests under consideration for various alternatives and for a selection of sample sizes. Power comparisons are then made in order to enable one to select the best test for any given alternative.

Finally, we consider a practical application and conclude with some general comments and recommendations.

3. TESTS AND THEIR NULL DISTRIBUTIONS

Let $X_1, X_2, \dots, X_n$ be a sequence of independent Bernoulli trials with success probabilities p_t and failure probabilities $q_t = 1 - p_t$, $t = 1, 2, \dots, n$. We shall focus on the distribution of $G_{n,k}$, the number of success runs of length at least k (for $k = 1, 2$ and 3) both under the null hypothesis

$$H_0: p_t = 0.5$$

and a general alternative hypothesis.

We developed a SAS computer program for the distribution of $G_{n,k}$,

under a general hypothesis, based directly on the theory given by Fu and Koutras (1994). However, these results can also be approximated with arbitrary accuracy by means of a simple simulation program.

The null distribution of $G_{n,k}$ was derived for $k = 1, 2, 3$ and for selected sample sizes n ($n \leq 40$).

Table 1 illustrates the case of $k = 3$, with the density function, $p(x) = P(G_{n,k} = x)$, given for all possible values of $G_{n,k}$ and rounded off to 3 decimal places.

TABLE 1. Null distribution of $G_{n,k}$ for $k = 3$ and selected values of n .

	x									
n	0	1	2	3	4	5	6	7	8	9
20	0.213	0.448	0.280	0.056	0.003	0.000				
30	0.092	0.303	0.361	0.193	0.046	0.005	0.000	0.000		
40	0.040	0.178	0.316	0.284	0.139	0.037	0.005	0.000	0.000	0.000

Inspection of the cases considered revealed some of the typical properties of the related test procedures. We found for small n and for $k > 1$, that the null distribution is positively skewed with mode close to 0. In such cases this skewness prevents one from obtaining critical values for left-sided testing when using a significance level $\alpha \leq 0.05$. This might necessitate the use of unrealistically high nominal significance levels when applied in left-sided testing.

4. POWER OF TESTS

The theory developed by Fu and Koutras (1994) was also used to calculate the distribution of runs test statistics under an alternative hypothesis. In the present study we formulated a variety of models representing different forms of nonrandomness. These models are all defined in terms of the success probability and given as follows:

Let $m(i, t) = \max\{2\text{mod}(t, i), 1\}$, $i = 3, 4$ and 5 with $t = 1, 2, \dots, n$. Also, let δ be a constant. $0 < \delta < 0.5$.

$$\text{Model 1: } p_t = 0.5 + \delta(-1)^{m(3,t)}$$

$$\text{Model 2: } p_t = 0.5 + \delta(-1)^{m(4,t)}$$

$$\text{Model 3: } p_t = 0.5 + \delta(-1)^{m(5,t)}$$

$$\text{Model 4: } p_t = 1 - 1/(2 + \delta t)$$

$$\text{Model 5: } p_t = 0.5 + \delta\{1 - (0.5)^t\}$$

$$\text{Model 6: } p_t = 0.5 - \delta(-1)^{m(3,t)}$$

$$\text{Model 7: } p_t = 0.5 - \delta(-1)^{m(4,t)}$$

$$\text{Model 8: } p_t = 0.5 - \delta(-1)^{m(5,t)}$$

$$\text{Model 9: } p_t = 0.5 + \delta(-1)^t$$

$$\text{Model 10: } p_t = 1/(2 + \delta t)$$

$$\text{Model 11: } p_t = 0.5 - \delta\{1 - (0.5)^t\}$$

Models 1–3, 6–9 represent situations where the success probability fluctuates (with varying frequencies) between a high and a low value, while models 4, 5, 10 and 11 represent either a monotonically increasing or monotonically decreasing sequence of success probabilities. Figure 1 gives a graphical representation of some of these models for the case of $\delta = 0.25$ and $n = 20$. Since some models are mirror images of others, only models 1–5 and 9 are represented in this figure.

When one tries to rationalize the model and the left- or right-sidedness for the three tests, one can observe the following from figure 1:

Model 1: A relatively large number of success runs of length at least 2 could be expected, so that $G_{n,2}$ should be applied on a right-sided basis.

Model 6: A relatively small number of success runs of length at least 2 could be expected, so that a left-sided $G_{n,2}$ or $G_{n,3}$ should be appropriate.

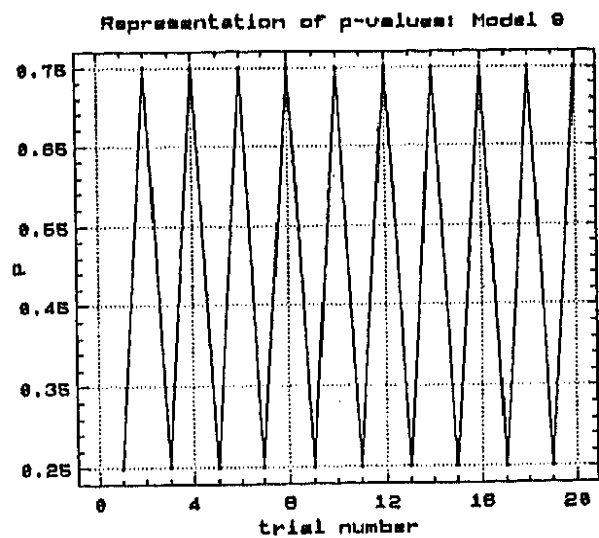
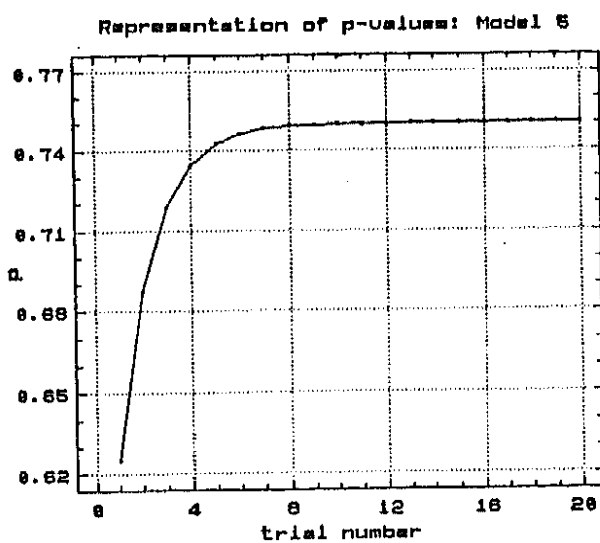
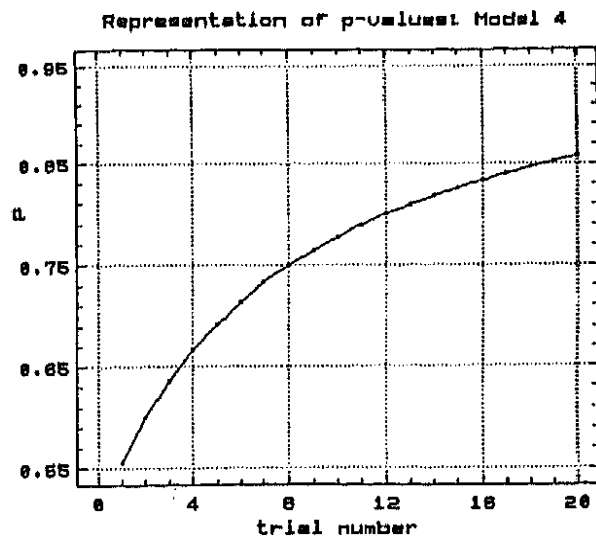
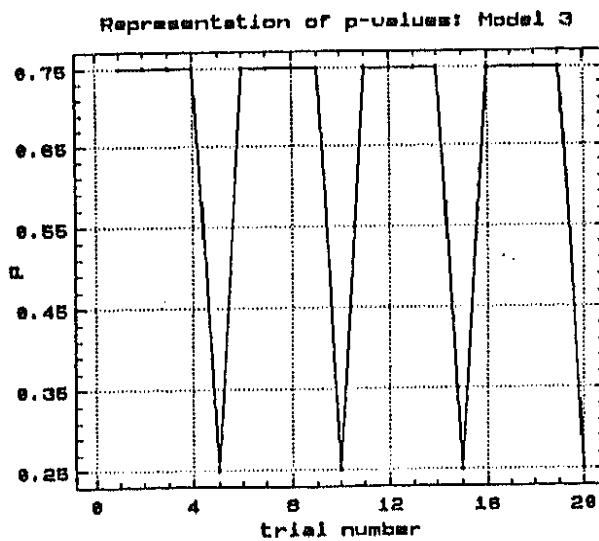
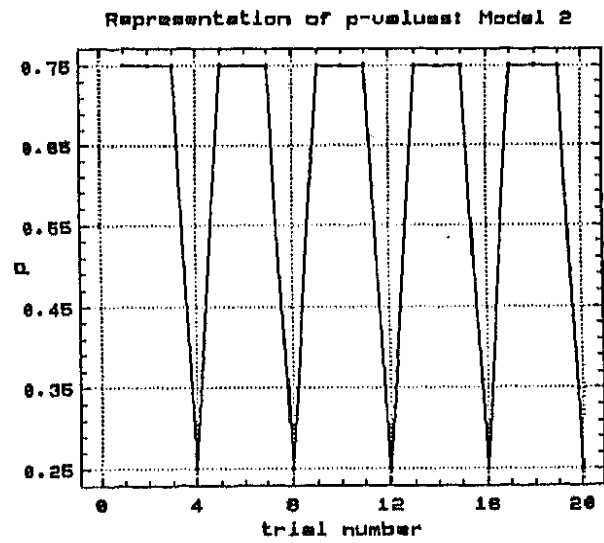
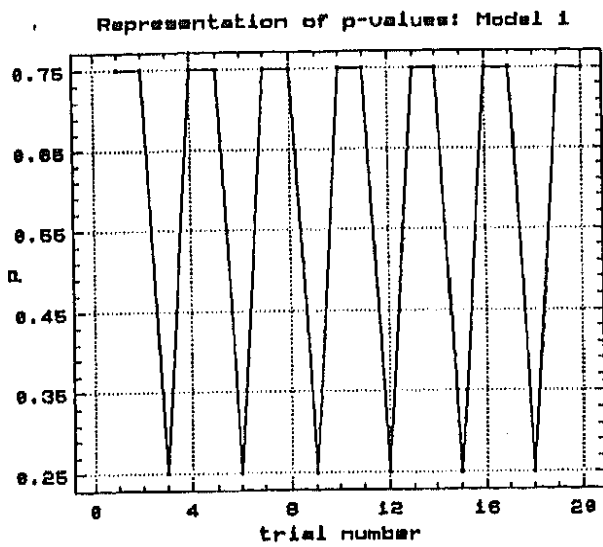


FIGURE 1. Graphical representation of 6 models of nonrandomness

Models 2 and 3: A large number of runs of length at least 3 could be expected, so that a right-sided $G_{n,3}$ should be applied.

Models 7 and 8: A small number of runs of length at least 2 could be expected, so that $G_{n,2}$ or $G_{n,3}$ should be used as left-sided procedures.

Model 9: A relatively large number of runs of length at least 1 could be expected, so that $G_{n,1}$ should be used on a right-sided basis.

Models 4 and 5: An increasing success probability should induce long success runs, so that left-sided runs tests should be applied.

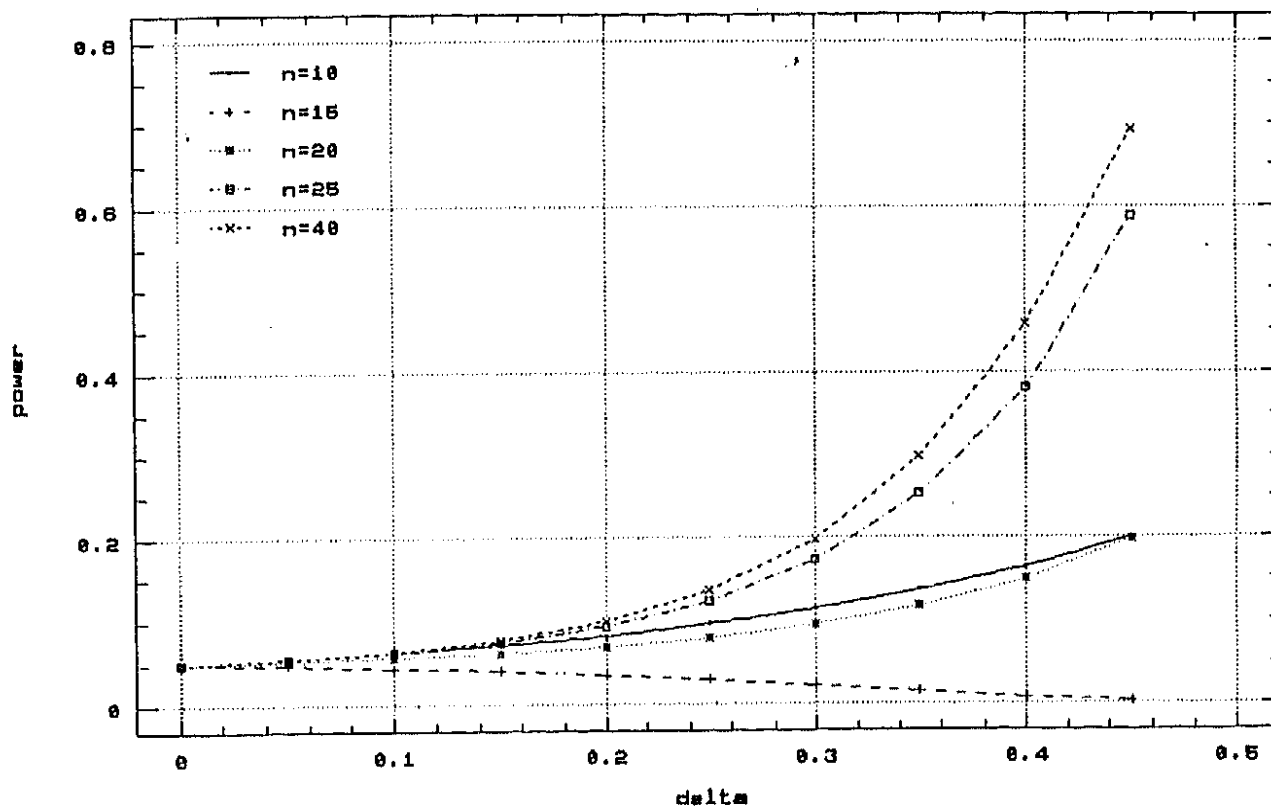
Models 10 and 11: A decreasing success probability could induce fewer and shorter success runs, so that one might expect tests on a left-sided basis to perform well.

In the power comparisons to follow, the value of the parameter δ was allowed to vary over the range 0.0 (0.05) 0.45. This enabled us to calculate the power function for 10 δ -values under each model. A significance level of 0.05 was used throughout. However, due to the positive skewness of the null distribution (as discussed in section 3 above), left-sided critical values had in certain cases to be chosen for levels set at larger multiples of 0.05. Randomization was used to enable us to compare the power of different tests at the same level.

4.1 Power comparison for fixed k and increasing n

Figures 2(a), 2(b) and 2(c) contain some examples of power functions of the test statistics $G_{n,k}$ for $k = 1, 2$ and 3 respectively. In all cases considered the power function was calculated for left-sided and right-sided testing, although for each case we reproduce the function only for either the left-sided or the right-sided test.

Power function of $G(n,k)$ for $k = 1$.
Model 1. Right-sided testing



Power function of $G(n,k)$ for $k = 1$.
Model 7. Left-sided testing

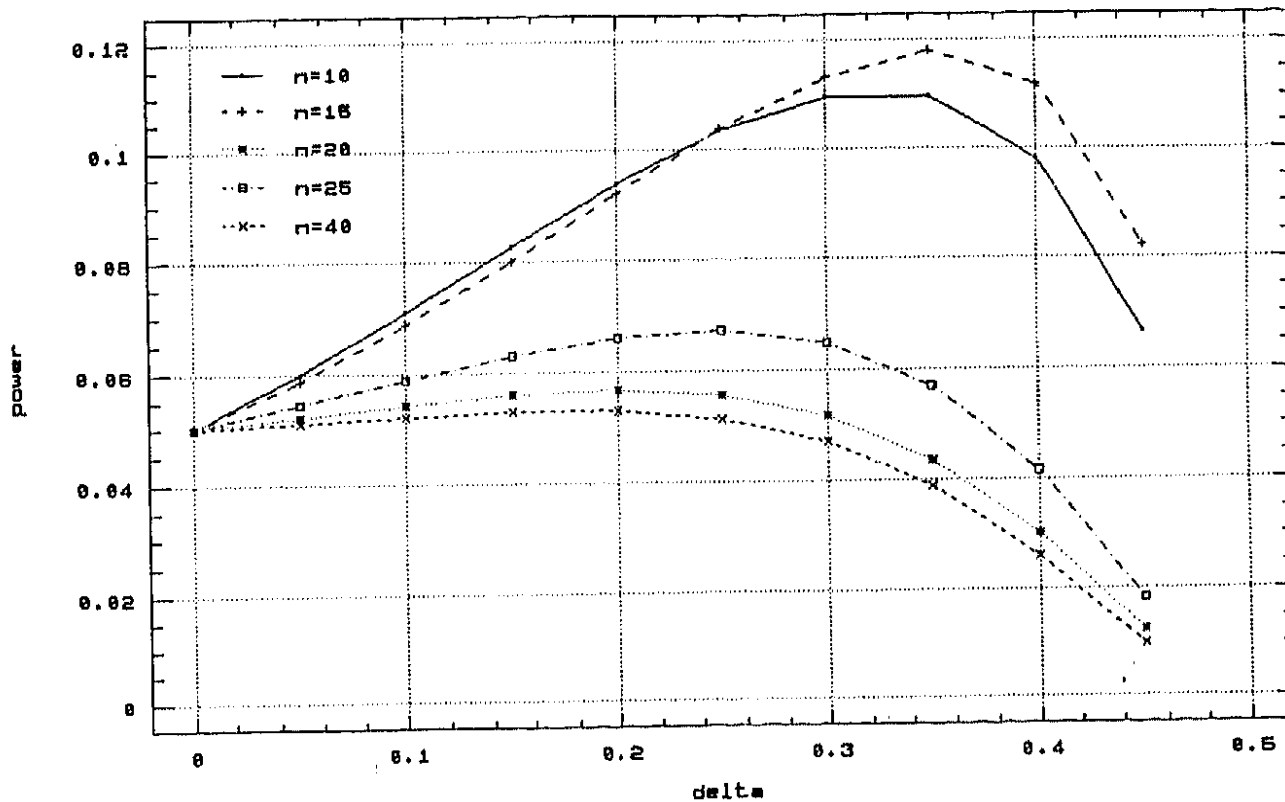
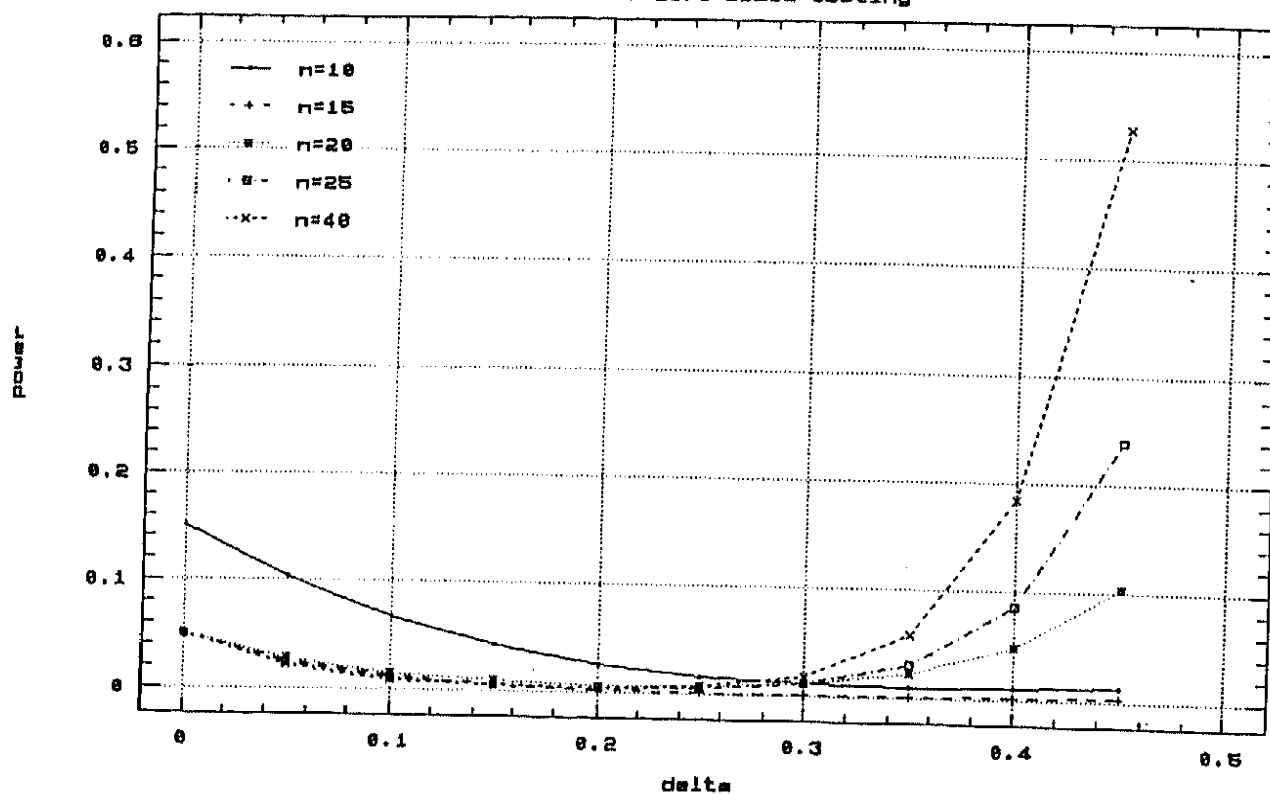


FIGURE 2(a). Power function of $G_{n,k}$ for $k = 1$ and selected models.

Power function of $G(n,k)$ for $k = 2$.
Model 6. Left-sided testing



Power function of $G(n,k)$ for $k = 2$.
Model 10. Left-sided testing

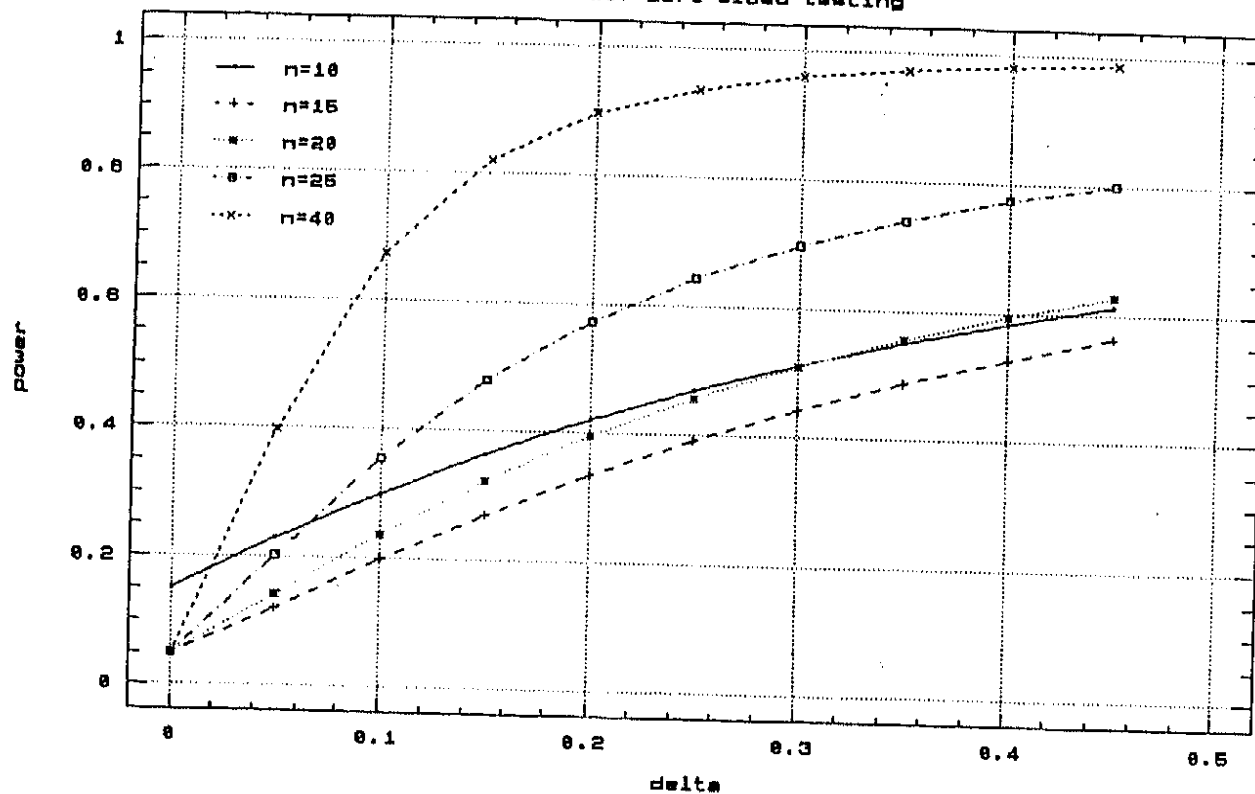


FIGURE 2(b). Power function of $G_{n,k}$ for $k = 2$ and selected models.

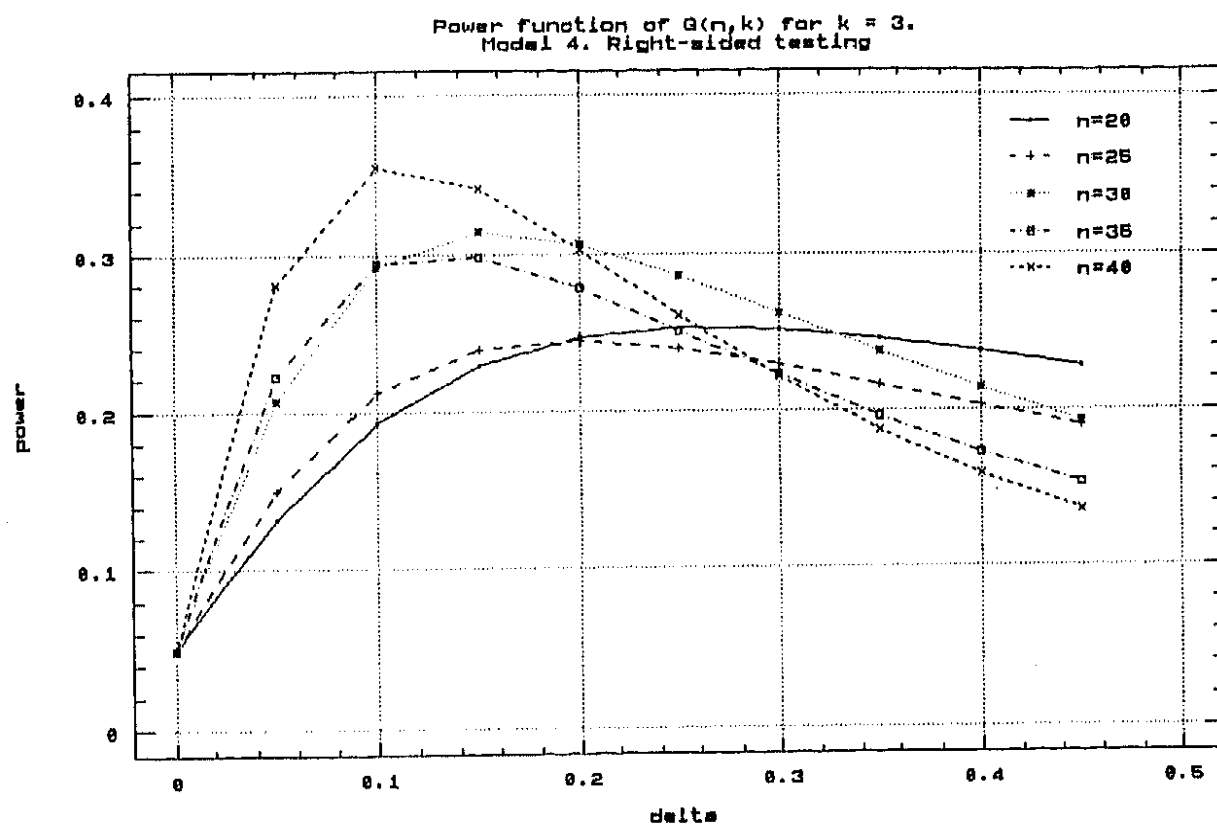
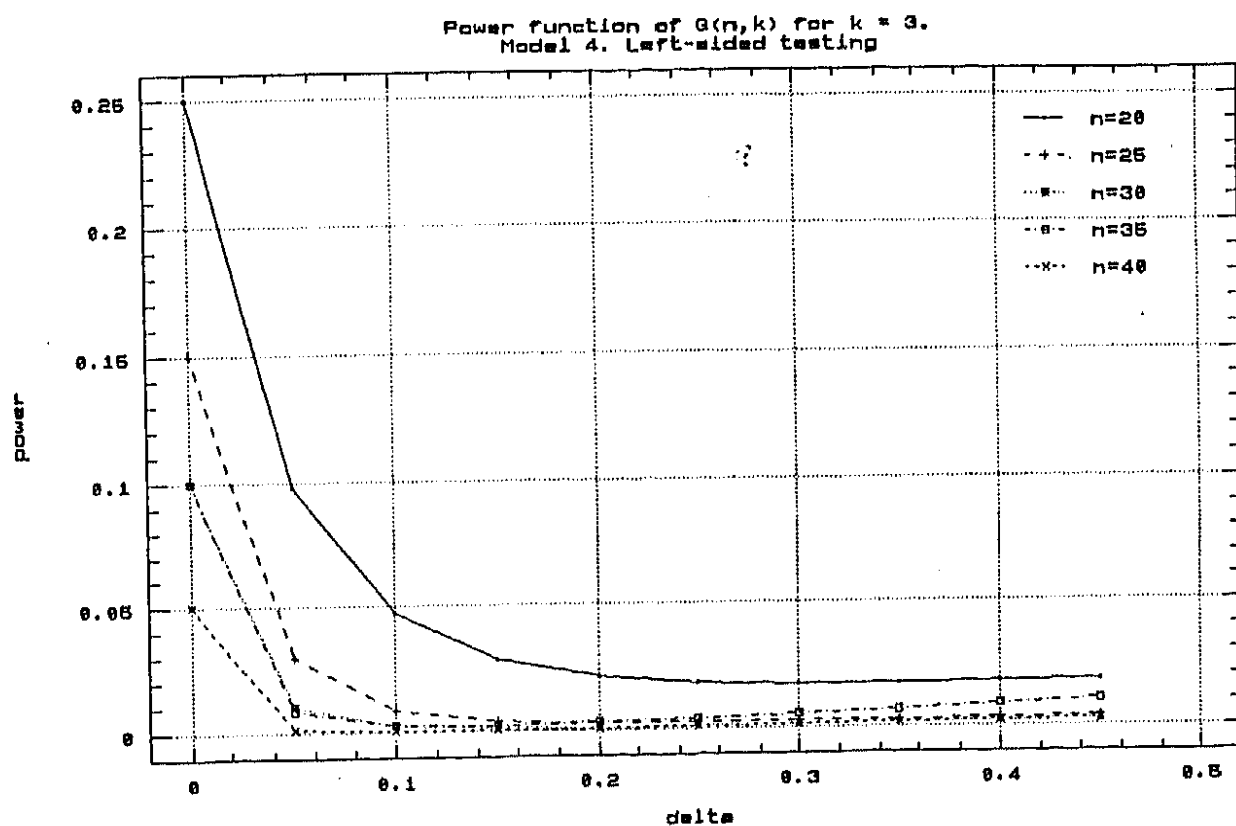


FIGURE 2(c). Power function of $G_{n,k}$ for $k = 3$ and selected models.

The tests mainly performed as expected for the different models (see discussion at the beginning of this section). There were some peculiar properties and shapes of the power functions and some of these are discussed below for each value of k :

$k = 1$:

Tests can be biased in some cases, since the power $\pi(\delta) < \alpha$, (the level) as seen for example in models 1 and 7. The power is not necessarily increasing in sample size n (see model 1).

The power function is not always monotonically increasing in the parameter δ (see model 7).

$k = 2$:

The power function can be decreasing for some n and increasing for another (see model 5).

The effect of the positive skewness of the null distribution of the test statistic is clearly seen in the power functions for models 5 and 10.

$k = 3$:

In some cases neither the left-sided nor the right-sided test is appropriate for the alternative under consideration. Also, the power function is not monotonically increasing in δ and it is also not consistently (for all δ) higher for larger n than for smaller n (see model 4, right-sided testing).

Most of the above-mentioned shortcomings in the power functions can be overcome by selecting the best test (that is, the correct choice of k) in each case. This will be discussed in the next section.

4.2 Power comparison for fixed n and different k

The exposition above leaves open the question of which test is the best in cases where 2 or 3 of them appear to perform well. For illustrative purposes

we decided to compare the procedures for $n = 40$ only. Figure 3 represents the power functions of $G_{40,k}$ for $k = 1, 2$ and 3 and the different models. A summary of the best choice of procedure for the different models is:

$G_{40,1}$: Models 4, 5 and 9.

$G_{40,2}$: Models 1, 7, 8, 10 and 11.

$G_{40,3}$: Models 2, 3, 6, 7 and 8.

As can be seen from the two graphs for model 5, given in figure 3, the right-sided $G_{40,3}$ performs better than the left-sided test. However, the peculiar shape of the power function of the right-sided $G_{40,3}$ needs special attention. A possible explanation is as follows: For smaller δ the success probability is higher than 0.5 but still leaves a reasonable chance for failures resulting in a sequence with relatively many runs of length at least 3 and consequently a higher probability of rejecting H_0 . For larger δ , the success probability is so large that the sequence consists mainly of successes resulting in few runs of length at least 3 and therefore a low probability of rejecting H_0 . From the two graphs it does not seem worthwhile considering a two-sided test either.

Looking at the comparison from a different angle, one can also conclude that:

$k = 1$ is the best choice or acceptable for models 4, 5, 9, 10 and 11;

$k = 2$ is that for models 1, 2, 3, 6, 7, 8, 10 and 11;

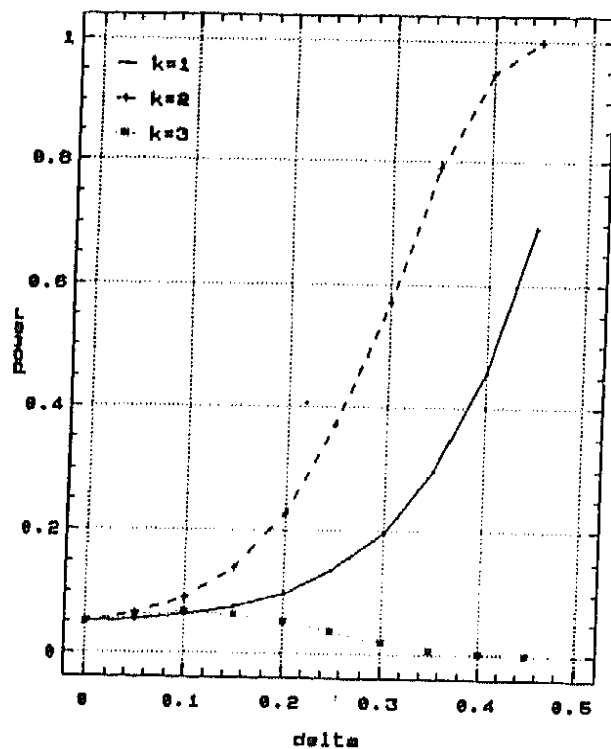
$k = 3$ is that for models 2, 3, 6, 7, 8, 10 and 11.

Finally, choices that should be avoided are:

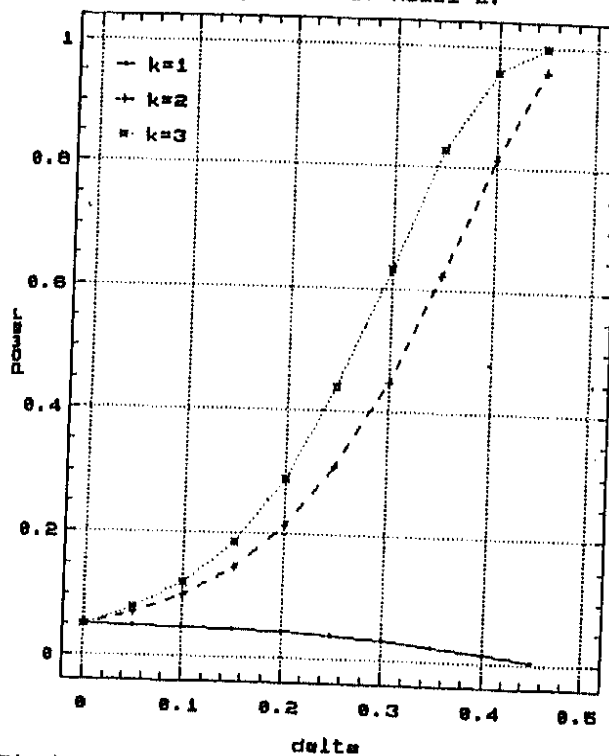
$k = 1$ for models 2, 3, 6, 7 and 8;

$k = 2$ for models 4, 5 and 9;

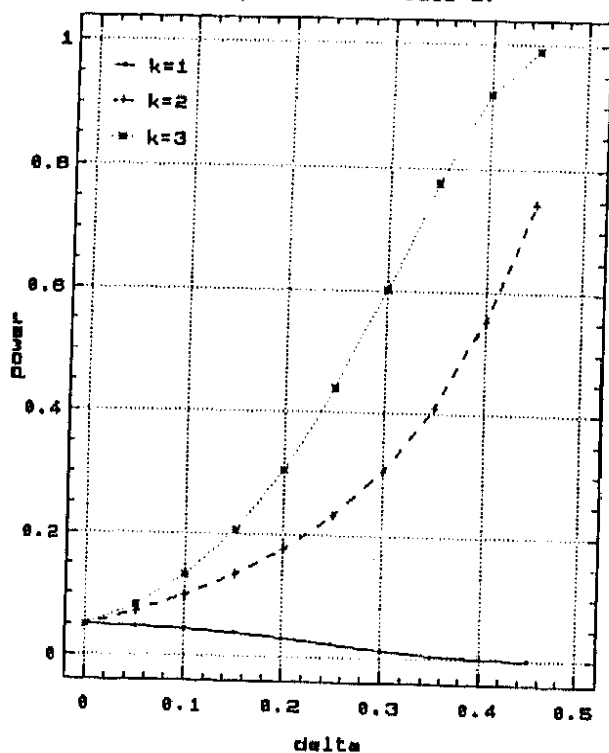
$k = 3$ for models 1, 4, 5 and 9.

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 . Model 1.

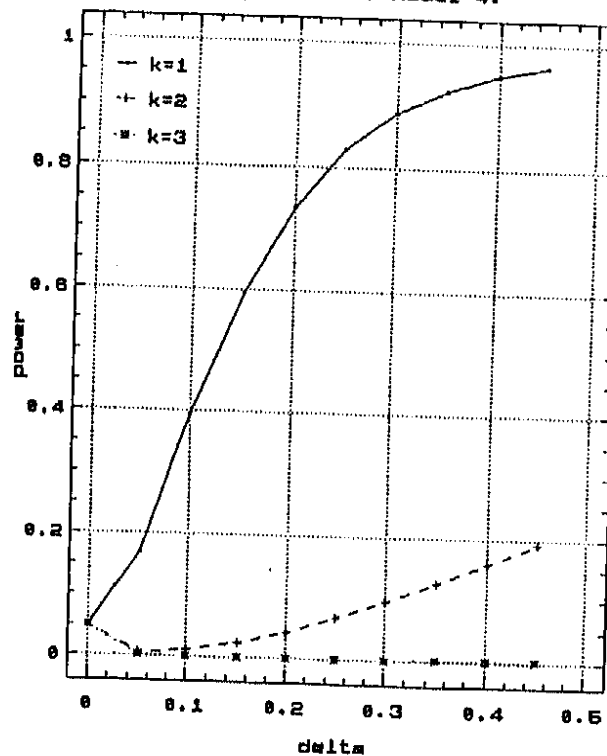
Right-sided testing

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 . Model 2.

Right-sided testing

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 . Model 3.

Right-sided testing

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 . Model 4.

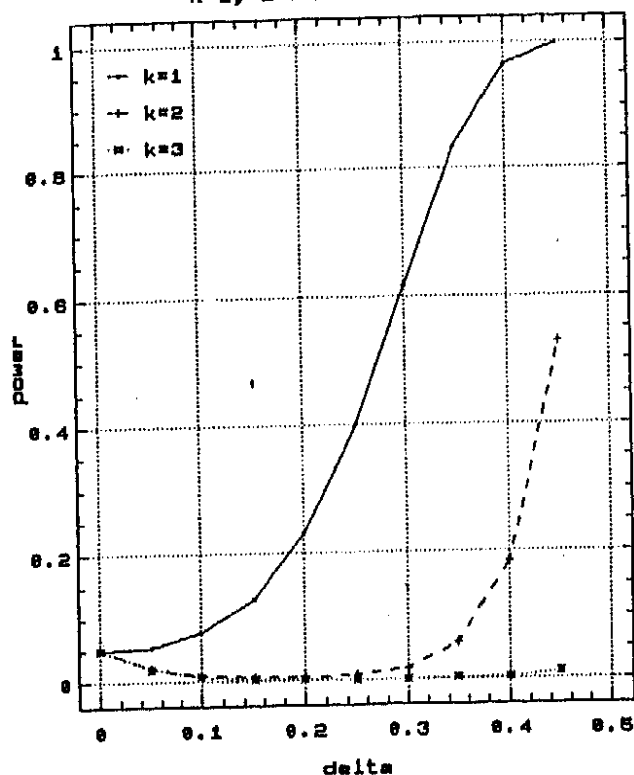
Left-sided testing

FIGURE 3. Comparison of power functions of $G_{40,k}$ for $k = 1, 2$ and 3 .

RUNS TESTS DISTRIBUTION

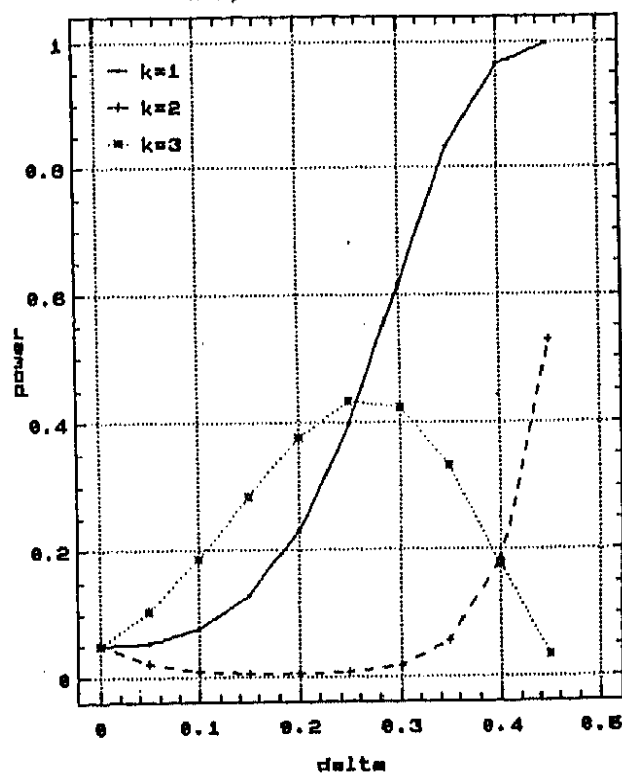
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Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 , Model 5.



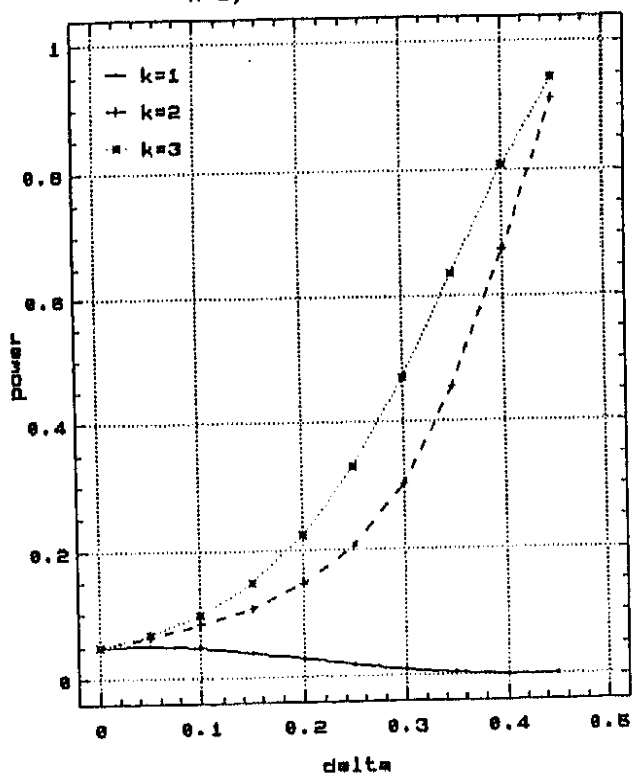
Left-sided testing

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 , Model 5.



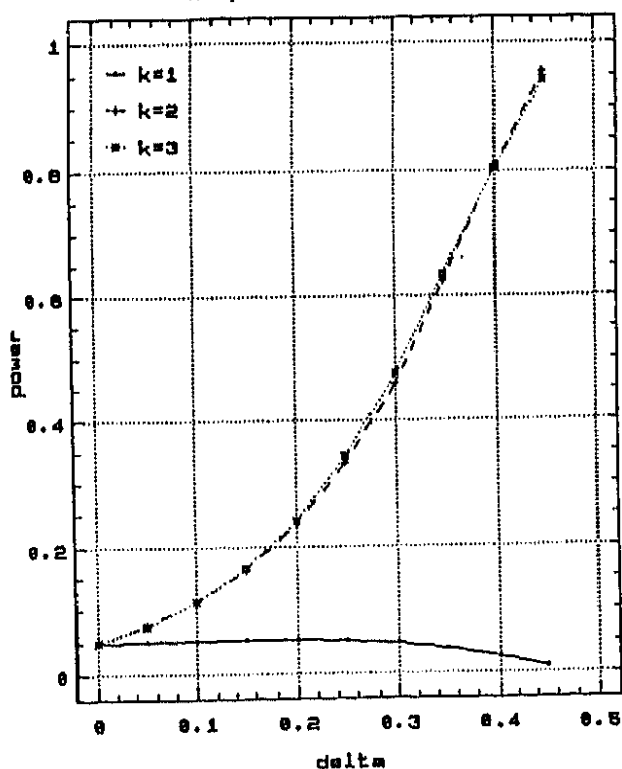
k=1, k=2 left and k=3 right-sided

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 , Model 6.



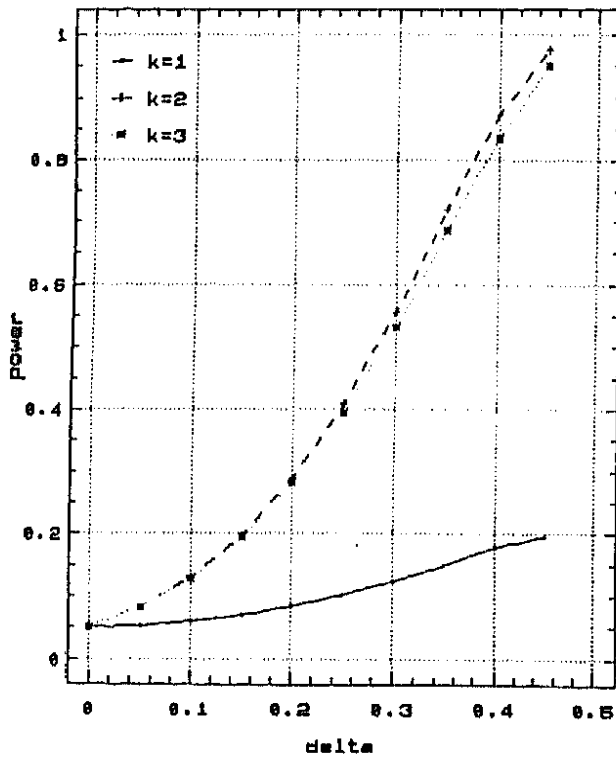
Left-sided testing

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 , Model 7.

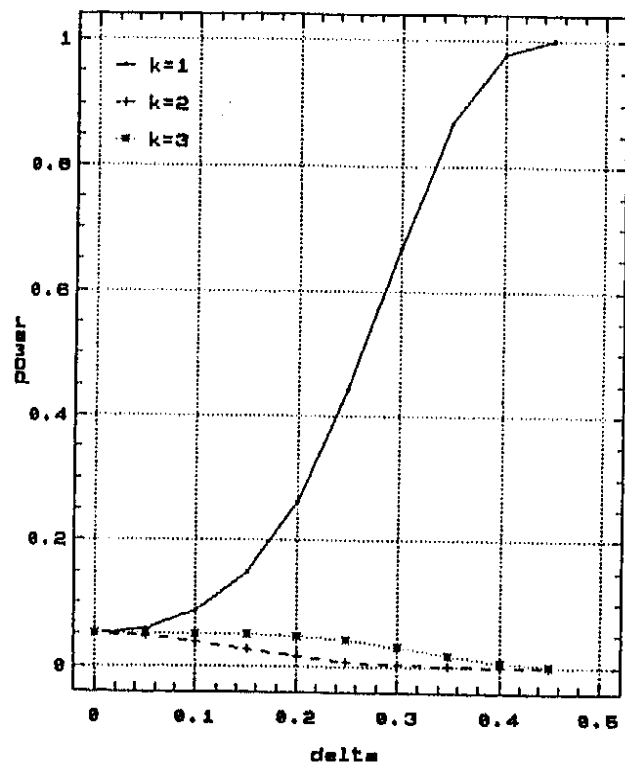


Left-sided testing

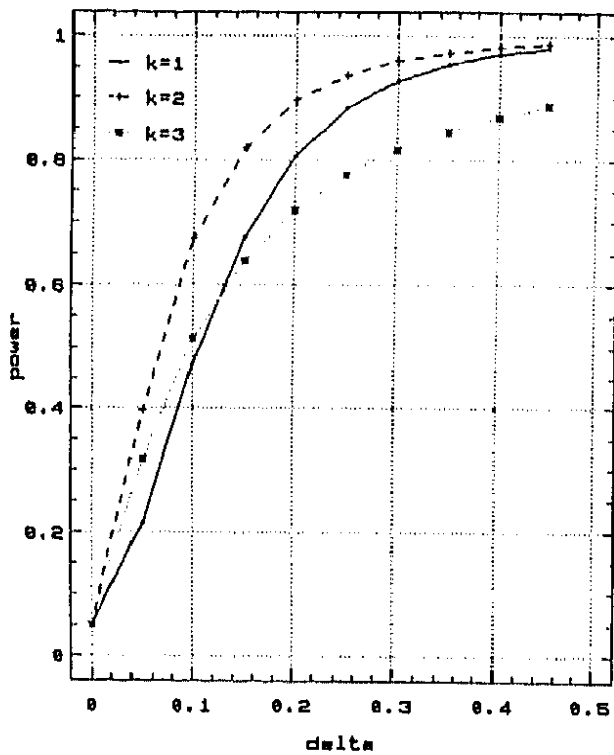
FIGURE 3. (continued)

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 . Model 8.

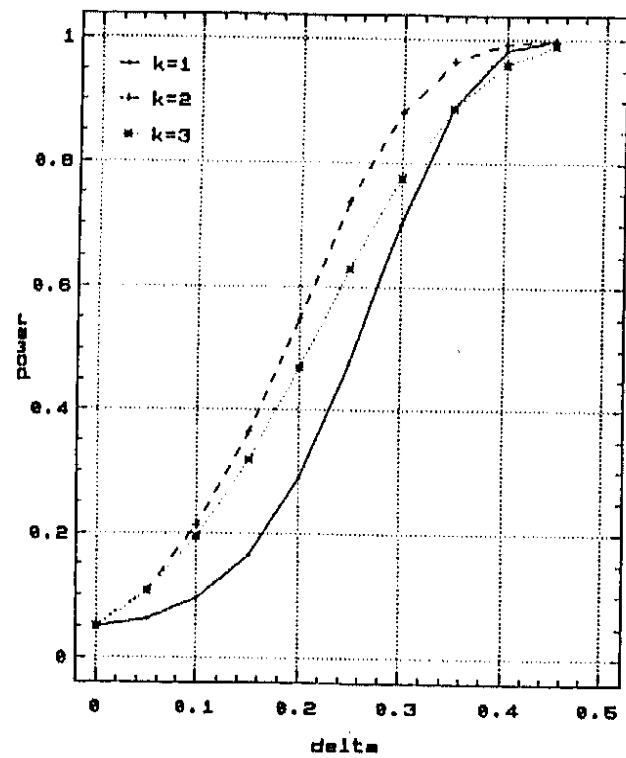
Left-sided testing

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 . Model 9.

Right-sided testing

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 . Model 10.

Left-sided testing

Power comparison of $G(n,k)$ for $n=40$ and $k=1, 2$ and 3 . Model 11.

Left-sided testing

FIGURE 3. (continued)

An important conclusion regarding the choice of the appropriate test is the availability of some knowledge (or some sound assumptions) about the underlying model. Even if this is available, one still has a choice between tests and some choices could be less favourable or even erroneous. A good example of this point is model 5 for which one would expect a left-sided $G_{n,k}$ to be best. The left-sided test based on $k = 1$ would indeed be best for large deviations from H_0 , but for small deviations, $k = 3$ would be better if right-sided testing is used (see figure 3).

5. APPLICATION

In the introduction we mentioned the use of runs tests for learning experiments in Psychology. As an application we consider determination of the appropriate choice of k in the class $G_{n,k}$ ($k = 1, 2$ and 3) for a problem of this type. Suppose we have a learning experiment where the subject behaves according to an underlying motive which can be described as "success breeds success and failure breeds failure". If p_t denotes success probability as before we can model this situation as follows: Let

$$\begin{aligned} P(\text{success in trial } t | \text{success in trial } t-1) &= p_{t-1}(1+a_t) \\ P(\text{success in trial } t | \text{failure in trial } t-1) &= p_{t-1}(1-b_t) \end{aligned}$$

where a_t and b_t are positive fractions with

$$a_t < (1 - p_{t-1})/p_{t-1}, \text{ for } t = 2, 3, \dots, n.$$

The series of success probabilities can be recursively written as follows:

$$p_t = p_{t-1}(1 - b_t) + p_{t-1}^2(a_t + b_t) \text{ for } t = 2, 3, \dots, n.$$

Taking $b_t = \delta$ and $a_t = \delta c(1 - p_{t-1})/p_{t-1}$, with $0 < \delta < 1$ and $0 < c < 1/\delta$, we can also write this as follows:

$$p_t = p_{t-1} \{1 + \delta(c - 1)q_{t-1}\} \text{ for } t = 2, 3, \dots, n$$

Therefore, p_t is increasing in t for $c > 1$ and decreasing in t for $c < 1$, where c is a constant which is related to the ratio of the success breeding force to the failure breeding force. In our application we choose $c = 1.5$. Finally, we use a sample size of 40 and allow δ to vary from 0.0 to 0.45 as before.

In figure 4 we represent the power functions $G_{n,k}$, $k = 1, 2$ and 3 for the best choices of left-sidedness or right-sidedness. We can see from this final figure that although $k = 1$ appeared to be the best choice for larger values of δ , $k = 3$ would have been better locally (in the neighbourhood of the null hypothesis).

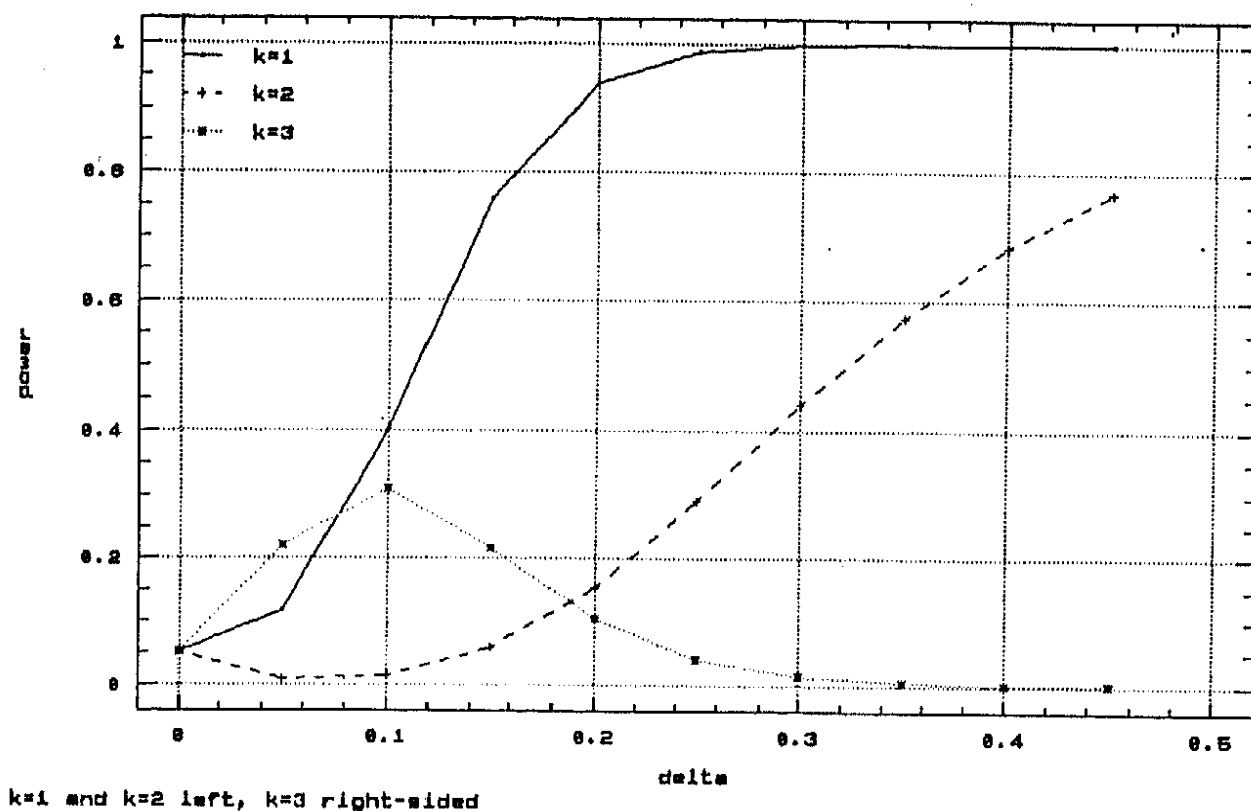


FIGURE 4. Power comparison of 3 runs tests for model "success breeds success and failure breeds failure"

6. CONCLUSION

As stated in the introduction, the theory developed by Fu and Koutras brings into reach a comprehensive study of runs test properties. This enabled us to develop a means of selecting the best test for a given alternative.

We then illustrated for a practical example from the psychology how to use this approach to find an appropriate member from the class of possible runs tests.

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